

SeedHammer II — the reasonably complex wallet

A four-tier degrading vault, taken end to end in **both** its wrappings: taproot and wsh. Six seeds in; two engraved backups out; the device agreeing with the host on every address, by two different routes.

Every command in this document ran, and its real stdout, stderr and exit code are reproduced. Every device screen is a frame the emulator rendered while an assertion held. Regenerate with `bash transcript_rcw.sh > transcript_rcw.txt` then `python3 build_pdf_rcw.py`; the builder refuses a transcript produced by a different version of its generator.

Test material.

The six seeds come from entropy 0x000...001 through 0x000...006 and the three passphrases are printed in the generator. Never put funds behind any of it.

What the starting representation actually is

The starting representation of "our reasonably complex wallet" is a BIP-388 WALLET POLICY TEMPLATE: a miniscript expression in which every key is a PLACEHOLDER -- @0 through @5 -- followed by that slot's derivation path and a <0;1> multipath suffix for the receive/change chains.

It is NOT a descriptor, NOT an address, and NOT a set of keys. It is the SHAPE of the wallet with key-shaped holes in it, and everything below is produced from it plus the six seeds. That distinction is the whole reason a keyless template can be engraved separately from the keys that fill it.

The four tiers, in the order the policy states them:

tier 1	@0 and @1 and @2 and sha256(H1)	-- at any time
tier 2	@3 and @4 and sha256(H2)	-- after older(32768), relative
tier 3	@5 alone	-- after height 1173520, absolute
tier 4	sha256(H3) alone	-- after height 1383520, absolute

Tier 4 has NO KEY. Whoever learns H3's preimage can spend it alone once that height passes, which is why every encode below carries --experimental and why the preimages are as sensitive as the seeds.

Two wrappings, which are DIFFERENT WALLETS with different addresses:

```
tr  taproot, NUMS internal key so there is no key-path spend; the four tiers
    are four separate taptree leaves. Keys at m/270028'/0'/8'/0'.
wsh  segwit v0; the taptree FLATTENS into one script, so the tiers become an
    or_i chain and multi_a becomes multi. Keys at m/270028'/0'/9'/1'.
```

The two policies

Same four tiers, two wrappings, and they are **different wallets**. The wsh form is not the tr form with a different prefix: multi_a becomes multi because CHECKSIGADD is tapscript-only, and the taptree — four separate leaves — flattens into one script, so the tiers become an or_i chain.

The consequence is worth stating: in tr a spend reveals only the leaf used. In wsh **every** spend reveals all four tiers, including the other two hashlock digests and both deadlines.

The two policies, as generated by derive-rcw-keys.sh:

```
$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-rcw/policy-tr.txt
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:sha256(edeb28a805feca09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi_a(3,@0/270028'/0'/8'/0'/<0;1>/*,@1/270028'/0'/8'/0'/<0;1>/*,@2/270028'/0'/8'/0'/<0;1>/*)),{and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,@3/270028'/0'/8'/0'/<0;1>/*,@4/270028'/0'/8'/0'/<0;1>/*))),{and_v(v:after(1173520),pk(@5/270028'/0'/8'/0'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954))}}})
[exit 0]

$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-rcw/policy-wsh.txt
wsh(or_i(and_v(v:sha256(edeb28a805feca09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi(3,@0/270028'/0'/9'/1'/<0;1>/*,@1/270028'/0'/9'/1'/<0;1>/*,@2/270028'/0'/9'/1'/<0;1>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi(2,@3/270028'/0'/9'/1'/<0;1>/*,@4/270028'/0'/9'/1'/<0;1>/*))),or_i(and_v(v:after(1173520),pk(@5/270028'/0'/9'/1'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954)))))
[exit 0]
```

Six seeds, six fingerprints

```
key-0: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
on abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
key-1: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
on abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
key-2: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
on abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
key-3: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
on abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
key-4: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
on abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
key-5: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
on abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon
```

TEST MATERIAL: entropy 0x000..001 through 0x000..006. Never put funds behind these.

key	fingerprint
@0	39ec1b6e
@1	d02bcedf
@2	5fd78eb3
@3	92c4c8b2
@4	086c047e
@5	cef8224c

6 distinct fingerprints across 6 seeds.

The master fingerprint is a property of the **seed**

, not of a derivation path, so it is identical in both wrappings. The generator asserts that rather than assuming it — if the two runs ever produced different fingerprints for one seed, something would be very wrong and silence would be the worst outcome.

The keys, and why 8 and 9

```
bg002h is m/270028'/coin'/account'/script', with the level-4 script value ruled
0' = tr and 1' = wsh. The operator selected ACCOUNT 8 for the tr form and
ACCOUNT 9 for the wsh form, so the two wallets hold disjoint keys derived from
the same six masters.

-- tr --
$ head -3 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-rcw/keys-tr/key-0.xpub
# @0 - tier 1 - 3-of-3 with hash A, spendable at any time
# tr wrapper; seed entropy 0x0000000000000000000000000000000000000000000000000000000000000001 (TEST ONLY)
# origin [39ec1b6e/270028'/0'/8'/0']
[exit 0]

-- wsh --
$ head -3 /scratch/code/shibboleth/mnemonic-engrave/design/journeys/inputs-rcw/keys-wsh/key-0.xpub
# @0 - tier 1 - 3-of-3 with hash A, spendable at any time
# wsh wrapper; seed entropy 0x0000000000000000000000000000000000000000000000000000000000000001 (TEST ONLY)
# origin [39ec1b6e/270028'/0'/9'/1']
[exit 0]

#####
##### tr
#####
```

tr — the cards

Fingerprints are declared (F-227): all six slots share one origin path, so without them a gathered key card matches every slot and seating must refuse the whole set.

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md encode tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547b
fee9ace803ac0,{and_v(v:sha256(ede0b28a805fec09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi_a(3,@0/270028'/0'/8'/
0'/<0;1>/*,@1/270028'/0'/8'/0'/<0;1>/*,@2/270028'/0'/8'/0'/<0;1>/*)),{and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc88
92441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,@3/270028'/0'/8'/0'/<0;1>/*,@4/270028'/0'/8'/0'/<0;1>/*))),{and_v
(v:after(1173520),pk(@5/270028'/0'/8'/0'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc270863
7dddf771c36d214056954))}})) --group-size 0 --experimental --fingerprint @0=39ec1b6e --fingerprint @1=d02bcedf --fingerprint
@2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerprint @4=086c047e --fingerprint @5=cef8224c
chunk-set-id: 0xec2a9
md1fas4fzqpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnmufuruzh27pyjw8n25ys5e6g4vx8jkqd
md1fas4fzqgx7jnay4pdl2kj98j9nyppq2z5e4cqqsqqye4m88qn2xlr50chjyfyk7t0l0yw9prny
md1fas4fzqjyr20txrg2rz78mwqmkran74xxmyryu7mk45szuz5e4kq3aaggz4fjuhtu99yrl935q
md1fas4fzqadsq9guvp636r6lzm8musa98376007chgv2yyenp8pp3hnh0hw8pk6gnah9vug436nmt
md1fas4fzppgptf2sxxjqu7cxmw8gzhnklf0a0r4nwfvfj9jsyxcpr7h80sgjvqqkycjn9c66qq9z
warning: --experimental relaxed the signature rule. This descriptor has at least one spend path that needs NO key, so whoever
r learns its preimage can spend it alone. If that preimage is engraved, THE PLATE IS BEARER ACCESS. Malleability, resource l
imits, repeated keys and timelock mixing were still checked.
note: stdout is a keyless descriptor template (no keys)
[exit 0]
```

The keyless template is 5 md1 chunk(s).

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md verify --template tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f
28ec96d547bfee9ace803ac0,{and_v(v:sha256(ede0b28a805fec09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi_a(3,@0/2700
28'/0'/8'/0'/<0;1>/*,@1/270028'/0'/8'/0'/<0;1>/*,@2/270028'/0'/8'/0'/<0;1>/*)),{and_v(v:older(32768),and_v(v:sha256(9ce09a8d
f1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,@3/270028'/0'/8'/0'/<0;1>/*,@4/270028'/0'/8'/0'/<0;1>
*))),{and_v(v:after(1173520),pk(@5/270028'/0'/8'/0'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a8
84ccc2708637dddf771c36d214056954))}})) --fingerprint @0=39ec1b6e --fingerprint @1=d02bcedf --fingerprint @2=5fd78eb3 --finge
rprint @3=92c4c8b2 --fingerprint @4=086c047e --fingerprint @5=cef8224c md1fas4fzqpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnmufuruz
h27pyjw8n25ys5e6g4vx8jkqd md1fas4fzqgx7jnay4pdl2kj98j9nyppq2z5e4cqqsqqye4m88qn2xlr50chjyfyk7t0l0yw9prny md1fas4fzqjyr20txrg
2rz78mwqmkran74xxmyryu7mk45szuz5e4kq3aaggz4fjuhtu99yrl935q md1fas4fzqadsq9guvp636r6lzm8musa98376007chgv2yyenp8pp3hnh0hw8pk6
gnah9vug436nmt md1fas4fzppgptf2sxxjqu7cxmw8gzhnklf0a0r4nwfvfj9jsyxcpr7h80sgjvqqkycjn9c66qq9z --experimental
OK
[exit 0]
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md inspect md1fas4fzqpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnmufuruz
h27pyjw8n25ys5e6g4vx8jkqd md1fas4fzqgx7jnay4pdl2kj98j9nyppq2z5e4cqqsqqye4m88qn2xlr50chjyfyk7t0l0yw9prny md1fas4fzqjyr20txrg
2rz78mwqmkran74xxmyryu7mk45szuz5e4kq3aaggz4fjuhtu99yrl935q md1fas4fzqadsq9guvp636r6lzm8musa98376007chgv2yyenp8pp3hnh0hw8pk6
gnah9vug436nmt md1fas4fzppgptf2sxxjqu7cxmw8gzhnklf0a0r4nwfvfj9jsyxcpr7h80sgjvqqkycjn9c66qq9z
... clipped; full text in transcript_rcw.txt
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md encode tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547b
fee9ace803ac0,{and_v(v:sha256(ede0b28a805fec09a77c48f82babc1249c79aa840de94fa4a85bf55a453c813),multi_a(3,@0/270028'/0'/8'/
0'/<0;1>/*,@1/270028'/0'/8'/0'/<0;1>/*,@2/270028'/0'/8'/0'/<0;1>/*)),{and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc88
92441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,@3/270028'/0'/8'/0'/<0;1>/*,@4/270028'/0'/8'/0'/<0;1>/*))),{and_v
(v:after(1173520),pk(@5/270028'/0'/8'/0'/<0;1>/*)),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc270863
7dddf771c36d214056954))}})) --key @0=xpub60dn2FdZCkR6b6nCFWKMZraCwEc7AedkRrVHF8DK5gSBrny6YJkiUzP9bL8j2a4osHSuHDsk1FBnSS81C82
qvZ4uiDofJGQKmNmxxqfUXCr --key @1=xpub60DqV8W2dzVkyfWmNUJHkwdH6JctbyGvKr3VUDUL1hSoUBGtFB3JtXoyYtvkVR356jyzagALehL56vqcjXAc
5ShoaCoL6gyrrtCLQlfdg2f --key @2=xpub6FMYX6RDPGTJgt3hYThGymmhRUhtuA9HagmHmDPUm1lhDKk6HLNfSpSyRWaqdiS9YMQUjjsL2Xn9ERJE9YDZR9jG
2X4UJb53XE3b8Z5Bw --key @3=xpub6E9jU3J1VtuNb5GVBatKMhYKHLMM7dD6yK2x4fhDD5ghXHLXJQgFZnwLEXWcljAb5ZCSNuHQGDqUdy3j2t3kRmSM6GFn
yBnM1PRXmD6bXb4 --key @4=xpub6Ep7XhWjqPzj5HTCQTf8M80RYUqGHJLXpQ9qbU7xJrGGCSG9Fba4QUtughDTBLJg5fSpraqaxKYUhmTvKMswiG8S24jJgL
zspG2tstkBdz --key @5=xpub6E6vqkLWC3QnMavFkCHB5K2fzKdH8FeikdWmuri09zyTWgNnwsZXf7KdcGAh9to36zwv0vdwsvr5hsxm2qf2h
qDuYuBW7q --fingerprint @0=39ec1b6e --fingerprint @1=d02bcedf --fingerprint @2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerpr
int @4=086c047e --fingerprint @5=cef8224c --group-size 0 --experimental
chunk-set-id: 0x30662
md1fxpnz8qpp2040veppfzzqqvpc4xdw7mc9j32q9lm9qnmufuruzh27pyjw8n25yph55lfqkpcz9ct5hzhvaa
md1fxpnz8q259ha26g57gzvzsyppg2nxxhqqzqqqnxhvuuzdgmuw3lz7g3jyjr20txrg2rz7q20ds83ekhpw3k
md1fxpnz8qnaqhdp7e12nrdjpw0dm26gppwp2v6mqgq7sy25e4kq4r3s828g0tugl0jqhfr09j9kxvddt
md1fxpnz8qe620r543swcz5gfnxwzr0hwlwuwrds5s4q454grrfq0vrdhr5ptem05hs6kpw79jrp43ync
md1fxpnz8p8tcavmjtzv3v5ppkqgl4emuzynqkvvgpp0kjhjdndqdhxe88vw6pfzLczdf76zqq8mkufmrwtcf
md1fxpnz8pve9q6stndtvgpmda6h50kpc65nezrrjpm94j9lfy6ns7qncgukzhpldcfwszqqw56mlrxmkpsjk
md1fxpnz8p3xw7332s5z7wjclvuel7ksrw329kvdzpqmetn093g7tezt7d9e9c664686ljgsynqk65spm59fc
md1fxpnz8pm0wgqppnnz2t6etcuqgw4cjs9jldx35ews7ldpzqu4eeegntc642f0fysud9jryjsxlwv9tvuptmt6
md1fxpnz8zq4cvqc5c66qznwuz8jvayr34rapk22tpkmrwxxyplpm0v28prczuppm3ewkk3js08d5zydcajz46
md1fxpnz8zqdr46wj9qdcfcuqn5d4mektc0jduuqxqg70f5hn7rp250u5sy7wjaftmvss39qt2k7n8ccqemc9
md1fxpnz8zh9cenn4kx57lte9spzutezsrdrj0kqkgzma9nuguk9zm882y22v6wuv0vdwsvr5hsxm2qf2h
md1fxpnz8zugfngxa4uzashfzfu063ryvvvrcwpy9echwgsdc4cvutz3692mfhe5d3l95lksccrfstkjzhkz2
md1fxpnz8rqz6umdpjgyfz5zqr9d2auqe8gpegqh97ejy8jvvqp7ekdlm9203gz5py08czsqaurwjghl5hs2
md1fxpnz8r060hu9umjh3rxzeuaz5u44smg3esd4vtt488c6k40mzcmdwz3g4gun7qx3s5xlrarft6uwdc
md1fxpnz8r597zah0lp6xaxlykwc49z08l3jk3hkj9tjgtc3kwjdjcu6t5aqqxpqu470znf8hs
warning: --experimental relaxed the signature rule. This descriptor has at least one spend path that needs NO key, so whoever
r learns its preimage can spend it alone. If that preimage is engraved, THE PLATE IS BEARER ACCESS. Malleability, resource l
imits, repeated keys and timelock mixing were still checked.
note: stdout is watch-only — public keys only, cannot spend
```

```
[exit 0]
```

The keyed card set is 15 md1 chunk(s).

```
wallet-descriptor-template-id (keyless): 68a1a888385797337ce5debc90fcfb1e
wallet-descriptor-template-id (keyed)  : 68a1a888385797337ce5debc90fcfb1e
wallet-policy-id                  (keyed) : 5c2c6fb54ce0c9042f9619dbe659472e
```

ONE WALLET, TWO IDS. The template id is key-STABLE and identical either way; the policy id depends on the keys. A card binds to whichever of the two matches its own form.

tr — descriptor and addresses

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md descriptor md1fxpnz8qpp2040veppfzzqqvpc4xdw7mc9j32q9l9qnmuf
ruz h27pyjw8n25yph55l1fqkpz9ct5hzhvaa md1fxpnz8q259ha26g57gzvsyypg2nxhqqqzqqqnxhvuuzdgmuw3lz7g3yjr20txrg2rz7q20ds83ekhpw3k md
1fxpnz8qnahqdmq7e12nrdjppjw0dm26gppwp2v6mqgq7sy25e4kqq4r3s828g0tugl0jqhfr09j9kxvddt md1fxpnz8q620ra57la3wscz5gfnxzwrr0hwlw
uwr5s5q454grrfqw0vrdrh5ptem05hs6kp79jrp43ync md1fxpnz8p8tcavmjtzv3v5ppkqgl4emuzynqkvvgp0kjhjdndqdhxe88vw6pfzlc2df76zqq8mkuf
mrwtcfnd md1fxpnz8pve9q6stndtvgpmda6h50kcp65nezrrjmp94j9lfy6ns7qncgukzhpldcfwszggw56mlrxmkpsjk md1fxpnz8p3xw7332s5z7wjclvuel7ks
rw329kvdzpqmetn093g7tezt7e9c664686ljgsynqk65spm59fc md1fxpnz8pm0wqgpnnzt6etcxuq4cjs9jldx35ews7l1dpzqu4eegntc642f0fysud9jryj
ryjsxlwv9tvuptmt6 md1fxpnz8zq4cvq5c66qznwuz8jvayr34rapk22tpkmrwxyp1pm0v28prczuppm3ewkk3js08d5zydcajz46 md1fxpnz8zdrq46wjqudcf
uqn5d4mektc0jduujxqg70f5hn7rpz50u5sy7wjaftmvss39qt2k7n8ccqemc9 md1fxpnz8zh9cenm4ksx7l7te9zpnstutezsrj0kgzgm9nugkut9zm882y2
2v6wuv0vddsvrs5hsx8m2qf2h md1fxpnz8zugfngxa4uzashfzf9u63ryvvryrcwpy9echwgsdc4cvutz3692mfhe5d3l95lkssecrftskjzhkzd md1fxpnz8r
qz6umdpjgyfz5qr9d2auqe8gpegqh97ejy8jvvqp7ekdlm9203gz25py08czsqaurwjghlh5hs2 md1fxpnz8r0960hu9xmjh3rxzufuaz5u44smg3esd4vttty48
8c6k40mzcmdwz3g4gun7qx3s5xlrrarft6uwdc md1fxpnz8r597zah0lp6xaxlykwc49z08l3jk3hkj9tjtg3kwdjcgub6t5aqqxpqu470znf8hs
tr(50929b74c1a04954b78b4b6035e97a5e078a5a0f28ec96d547bfee9ace803ac0,{and_v(v:sha256(ede0b28a805feca09a77c48f82babcb1249c79aa8
40de94fa4a85bf55a453c813)},multi_a(3,{39ec1b6e/270028'/0'/8'/0'}xpub661MyMwAQBcEnD2AtBLZFdpjyX4eUjLYHGbaJM5uUaNHlMgaovDDvj6k
jxx987aSzYkBaAo6nsse35VZ4hEW3ya1R1Kj5oSiMEzmYHYCYH/<0;1>/*,[d02bcdcf/270028'/0'/8'/0']xpub661MyMwAQBcF1ymCBSeBwHofzVWnLXr3p
L4oc8CBveucrmMCKS0wxcU0Uq0cP9A7JRwm5Aq5W8DEXGJjXBDX76ke8XQ6aVtAJP7HX/<0;1>/*,[5fd78eb3/270028'/0'/8'/0']xpub661MyMwAQBc
EnXhS9XNh1lLetXt6lkvxw1eYtjtrHYrysh3wyxoxUaNXG8TDehBtM3J5R7CHnZS6gdT8PY3kY3PJqQCYriq4swVEAg57/<0;1>/*)},and_v(v:older(32
768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi_a(2,{92c4c8b2/270028'/0'/8'/0'}x
pub661MyMwAQBcEdS8bQ28dKzUuYqJkaMomhRdmC84fXrrsZBWAwJxCFFDsVLfQn8HdBXqNMCAgHKOzqqgy6aDKcVMYsjrhNYLxp49VeJj8y/<0;1>/*,[086c
047e/270028'/0'/8'/0']xpub661MyMwAQBcEsTQ3dWNcXrFdaVqoMP4FN1HgB4vHqRbXABdmsTKHLX2J4nJiDwExnr32u7sX5KstfjUq3LmzteztWRCmpdtT3
FBNX7Xzv5/<0;1>/*)},and_v(v:after(1173520),pk([cef8224c/270028'/0'/8'/0']xpub661MyMwAQBcF68jPjC8VwRUTZfaqPMWx2f3RT7TBH
1hXzC6rYwDNyevkEj49rowooqTSvqQzmC3z7gkD6547TYCHZ1MMcwSsmktWDx7/<0;1>/*)},and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3
dfec5d0c0a884ccc2708637d3dd7f71c36d214056954))))}#h64g8jju
note: stdout is watch-only – public keys only, cannot spend
[exit 0]
```

1299 bytes

Five receive and five change addresses, on the host (5 and 5 derived):

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md address md1fxpnz8qpp2040veppfzzqqvpc4xdw7mc9j32q9l9qnmufruz
h27pyjw8n25yph55l1fqkpz9ct5hzhvaa md1fxpnz8q259ha26g57gzvsyypg2nxhqqqzqqqnxhvuuzdgmuw3lz7g3yjr20txrg2rz7q20ds83ekhpw3k md1fx
pnz8qnahqdmq7e12nrdjppjw0dm26gppwp2v6mqgq7sy25e4kqq4r3s828g0tugl0jqhfr09j9kxvddt md1fxpnz8q620ra57la3wscz5gfnxzwrr0hwlwuwr
d5s5q454grrfqw0vrdrh5ptem05hs6kp79jrp43ync md1fxpnz8p8tcavmjtzv3v5ppkqgl4emuzynqkvvgp0kjhjdndqdhxe88vw6pfzlc2df76zqq8mkufmrw
tcfn md1fxpnz8pve9q6stndtvgpmda6h50kcp65nezrrjmp94j9lfy6ns7qncgukzhpldcfwszggw56mlrxmkpsjk md1fxpnz8p3xw7332s5z7wjclvuel7ks
rw329kvdzpqmetn093g7tezt7e9c664686ljgsynqk65spm59fc md1fxpnz8pm0wqgpnnzt6etcxuq4cjs9jldx35ews7l1dpzqu4eegntc642f0fysud9jryj
sx1wv9tvuptmt6 md1fxpnz8zq4cvq5c66qznwuz8jvayr34rapk22tpkmrwxyp1pm0v28prczuppm3ewkk3js08d5zydcajz46 md1fxpnz8zdrq46wjqudcf
uqn5d4mektc0jduujxqg70f5hn7rpz50u5sy7wjaftmvss39qt2k7n8ccqemc9 md1fxpnz8zh9cenm4ksx7l7te9zpnstutezsrj0kgzgm9nugkut9zm882y2
2v6wuv0vddsvrs5hsx8m2qf2h md1fxpnz8zugfngxa4uzashfzf9u63ryvvryrcwpy9echwgsdc4cvutz3692mfhe5d3l95lkssecrftskjzhkzd md1fxpnz8r
qz6umdpjgyfz5qr9d2auqe8gpegqh97ejy8jvvqp7ekdlm9203gz25py08czsqaurwjghlh5hs2 md1fxpnz8r0960hu9xmjh3rxzufuaz5u44smg3esd4vttty488c6
k40mzcmdwz3g4gun7qx3s5xlrrarft6uwdc md1fxpnz8r597zah0lp6xaxlykwc49z08l3jk3hkj9tjtg3kwdjcgub6t5aqqxpqu470znf8hs --chain 0 --c
ount 5
bc1p8h77a4mg149ml6d6crdpvx4j5g6gclkv9vr93en154cdz0gp6v8smuqr2n
bc1panm4lv4hl8n96xccufj7c7rkd4hhuvled2hp5eqdhncavpuvfdq5z9t8t
bc1pf0j39ep28fqsssev4xf6mrtjnp5c896gqklwu5tw2lvqcvg7jzxxs63pwxu
bc1ps2533c3n6agpt48lr3x9ugk77zlfctfug03upeh74fzjtchfypqnnljqn
bc1prf5sadb5390xv2n0famzsc836k8alxqq9r7sfzxyng6qzrpp63sw9ym0w
note: stdout is watch-only – public keys only, cannot spend
[exit 0]
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md address md1fxpnz8qpp2040veppfzzqqvpc4xdw7mc9j32q9l9qnmufruz
h27pyjw8n25yph55l1fqkpz9ct5hzhvaa md1fxpnz8q259ha26g57gzvsyypg2nxhqqqzqqqnxhvuuzdgmuw3lz7g3yjr20txrg2rz7q20ds83ekhpw3k md1fx
pnz8qnahqdmq7e12nrdjppjw0dm26gppwp2v6mqgq7sy25e4kqq4r3s828g0tugl0jqhfr09j9kxvddt md1fxpnz8q620ra57la3wscz5gfnxzwrr0hwlwuwr
d5s5q454grrfqw0vrdrh5ptem05hs6kp79jrp43ync md1fxpnz8p8tcavmjtzv3v5ppkqgl4emuzynqkvvgp0kjhjdndqdhxe88vw6pfzlc2df76zqq8mkufmrw
tcfn md1fxpnz8pve9q6stndtvgpmda6h50kcp65nezrrjmp94j9lfy6ns7qncgukzhpldcfwszggw56mlrxmkpsjk md1fxpnz8p3xw7332s5z7wjclvuel7ks
rw329kvdzpqmetn093g7tezt7e9c664686ljgsynqk65spm59fc md1fxpnz8pm0wqgpnnzt6etcxuq4cjs9jldx35ews7l1dpzqu4eegntc642f0fysud9jryj
sx1wv9tvuptmt6 md1fxpnz8zq4cvq5c66qznwuz8jvayr34rapk22tpkmrwxyp1pm0v28prczuppm3ewkk3js08d5zydcajz46 md1fxpnz8zdrq46wjqudcf
uqn5d4mektc0jduujxqg70f5hn7rpz50u5sy7wjaftmvss39qt2k7n8ccqemc9 md1fxpnz8zh9cenm4ksx7l7te9zpnstutezsrj0kgzgm9nugkut9zm882y2
2v6wuv0vddsvrs5hsx8m2qf2h md1fxpnz8zugfngxa4uzashfzf9u63ryvvryrcwpy9echwgsdc4cvutz3692mfhe5d3l95lkssecrftskjzhkzd md1fxpnz8r
qz6umdpjgyfz5qr9d2auqe8gpegqh97ejy8jvvqp7ekdlm9203gz25py08czsqaurwjghlh5hs2 md1fxpnz8r0960hu9xmjh3rxzufuaz5u44smg3esd4vttty488c6
k40mzcmdwz3g4gun7qx3s5xlrrarft6uwdc md1fxpnz8r597zah0lp6xaxlykwc49z08l3jk3hkj9tjtg3kwdjcgub6t5aqqxpqu470znf8hs --chain 1 --c
ount 5
bc1pt7g96plv4mw73vrs64wvq2eswq9zrh4r0ztnx7huuma94xzsckhkq8vkn4
bc1pp3fpcxm9zkp75ghuhj5qh804wxzrf5tp6wcunr0up7j3xp46v3su0mntq
bc1pg6urxdm50ujxw8kl7zj17d2nwh75hx14hphjalzxx7pv2ewuy9qccund
bc1pf8rf0ylce0g0dd4k44w055q4jnywsh7mnd7ysnqt4xmy9fqat6ws0l38ps
bc1p80t9jzls9624gttgq8tw65q7jlvk395679qh77pj3quv02uu3dq6xm86k
note: stdout is watch-only – public keys only, cannot spend
[exit 0]
```


wsh — the cards

Fingerprints are declared (F-227): all six slots share one origin path, so without them a gathered key card matches every slot and seating must refuse the whole set.

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md encode wsh(or_i(and_v(v:sha256(ede0b28a805feca09a77c48f82bab
1249c79aa840de94fa4a85bf55a453c813),multi(3,@0/270028'/0'/9'/1'/<0;1>/*,@1/270028'/0'/9'/1'/<0;1>/*,@2/270028'/0'/9'/1'/<0;1
>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi(2,@3/
270028'/0'/9'/1'/<0;1>/*,@4/270028'/0'/9'/1'/<0;1>/*))),or_i(and_v(v:after(1173520),pk(@5/270028'/0'/9'/1'/<0;1>/*))),and_v
(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954)))))) --group-size 0 --experimenta
l --fingerprint @0=39ec1b6e --fingerprint @1=d02bcdcf --fingerprint @2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerprint @4=0
86c047e --fingerprint @5=cef8224c
chunk-set-id: 0xf2bf0
md1f72lszqpp2040veppfx8qqpx2v6aahst9z5qtlk2pxnhcj8c9w4uzfyu0x4ggmed54w92q28q4
md1f72lszqgda986f2zm74dy20yppxxzq5efntsqppqqfntkwwpx5d78gl30ygjgsle764dhvdvgr
md1f72lszq5gx57kvxs5x9u0kuph8m8a2vdkgxfeahdtgcy9cefntvqpr6qs92n9l9npextjdsvk
md1f72lszq6mq23ccr4r5847y0hep620ra57la3wscz5gfnxzwzrr0hlwuwrds5spa2we2d8dteg
md1f72lszpzszkj4qvdypeasdkuws908d7jl678txujcny9qgdsz8awwl3ycqcn0sdc05cheg5a
warning: --experimental relaxed the signature rule. This descriptor has at least one spend path that needs NO key, so whoever
r learns its preimage can spend it alone. If that preimage is engraved, THE PLATE IS BEARER ACCESS. Malleability, resource l
imits, repeated keys and timelock mixing were still checked.
note: stdout is a keyless descriptor template (no keys)
[exit 0]
```

The keyless template is 5 md1 chunk(s).

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md verify --template wsh(or_i(and_v(v:sha256(ede0b28a805feca09a7
7c48f82bab1249c79aa840de94fa4a85bf55a453c813),multi(3,@0/270028'/0'/9'/1'/<0;1>/*,@1/270028'/0'/9'/1'/<0;1>/*,@2/270028'
/0'/9'/1'/<0;1>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76a
d),multi(2,@3/270028'/0'/9'/1'/<0;1>/*,@4/270028'/0'/9'/1'/<0;1>/*))),or_i(and_v(v:after(1173520),pk(@5/270028'/0'/9'/1'/<0;
1>/*))),and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954)))))) --fingerprint @0
=39ec1b6e --fingerprint @1=d02bcdcf --fingerprint @2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerprint @4=086c047e --fingerpr
int @5=cef8224c md1f72lszqpp2040veppfx8qqpx2v6aahst9z5qtlk2pxnhcj8c9w4uzfyu0x4ggmed54w92q28q4 md1f72lszqgda986f2zm74dy20ypp
xxzq5efntsqppqqfntkwwpx5d78gl30ygjgsle764dhvdvgr md1f72lszq5gx57kvxs5x9u0kuph8m8a2vdkgxfeahdtgcy9cefntvqpr6qs92n9l9npextj
dsvk md1f72lszq6mq23ccr4r5847y0hep620ra57la3wscz5gfnxzwzrr0hlwuwrds5spa2we2d8dteg md1f72lszpzszkj4qvdypeasdkuws908d7jl678t
xujcny9qgdsz8awwl3ycqcn0sdc05cheg5a --experimental
OK
[exit 0]
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md inspect md1f72lszqpp2040veppfx8qqpx2v6aahst9z5qtlk2pxnhcj8c9
w4uzfyu0x4ggmed54w92q28q4 md1f72lszqgda986f2zm74dy20yppxxzq5efntsqppqqfntkwwpx5d78gl30ygjgsle764dhvdvgr md1f72lszq5gx57kvxs
5x9u0kuph8m8a2vdkgxfeahdtgcy9cefntvqpr6qs92n9l9npextjdsvk md1f72lszq6mq23ccr4r5847y0hep620ra57la3wscz5gfnxzwzrr0hlwuwrds
5spa2we2d8dteg md1f72lszpzszkj4qvdypeasdkuws908d7jl678txujcny9qgdsz8awwl3ycqcn0sdc05cheg5a
... clipped; full text in transcript_rcw.txt
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md encode wsh(or_i(and_v(v:sha256(ede0b28a805feca09a77c48f82bab
1249c79aa840de94fa4a85bf55a453c813),multi(3,@0/270028'/0'/9'/1'/<0;1>/*,@1/270028'/0'/9'/1'/<0;1>/*,@2/270028'/0'/9'/1'/<0;1
>/*)),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18bc7db81bb0fb3f54c6d9064e7b76ad),multi(2,@3/
270028'/0'/9'/1'/<0;1>/*,@4/270028'/0'/9'/1'/<0;1>/*))),or_i(and_v(v:after(1173520),pk(@5/270028'/0'/9'/1'/<0;1>/*))),and_v
(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637dddf771c36d214056954)))))) --key @0=xpub6EddmgK6uMbSt72
51zES359MJzfz3o2wTJweeffzSiPJf1FUG6easA2UK3jUbztBffwS4Dg20WjhomU2jJKr4EZukDfxVxb4VJva3jXGAK --key @1=xpub6EPimu5ztRjy6dFEcV
b9AQNAadNHN8qumZyQHwPwH9U9xJ9XMi328aXAFMGHwc1lwZWrmdwsHUPWJn5in6BHyw6rBj3X5ZLUWe8xhNPRf5kL --key @2=xpub6EVA2mZiCBtGyBGM5eEs4
2k8ts04QekUog2t8U14UDyjiUYpKBYLYcXB8t3yVAvapQymRSK2MDM84csPZa5u0QEUESVG028RJt47eeC846C --key @3=xpub6DZrOuPBj8Z5JCj4DP2t3Cf
D8vESCKb1Y4C6lGqU364QK3tRVUfufUERM8FGiFHXGKSNEALDMXgDJzVKDg79usEoB8F56Tf3gyNFQ4TjrHq --key @4=xpub6FFo9cyULG6UEz8usJ7R8JQjvwE
sjaNCSK4jv2WUxsFJm4TPXR6zUpdsKg4MoFE179in7ilqKFKJ1CswEWSntz2FW2tLYSiYluYpjh9ataZa --key @5=xpub6DjFP13AXC9SPE6XnxQ2SKsF15r1K
bjcNUEkfABA3WAmGfKECo8HNqfU71dmPoQshBGqzubyYhvgiSF44qHUh64WnM8Z42qdf2fDKMCP57eX --fingerprint @0=39ec1b6e --fingerprint @1=d02
bcdcf --fingerprint @2=5fd78eb3 --fingerprint @3=92c4c8b2 --fingerprint @4=086c047e --fingerprint @5=cef8224c --group-size 0
--experimental
chunk-set-id: 0xa4ff4
md1f5n158qpp2040veppfx8qqpx2v6aahst9z5qtlk2pxnhcj8c9w4uzfyu0x4ggr0ff7jqeh2qcmjtmnm68
md1f5n158qdg06453fusycggzn9xdwqqyqqpxdweecy63hcar79u3zfygx57kvxs5x9uqf93jqxfl9ppns
md1f5n158qhmwmqkran74xmyryu7mk45vzzuv5e4kq3aqg4fntvqp28rqw5s7hc37lyqg66puucsn7wgk
md1f5n158qm5578fmlzaps9gdsnxvyuyxlla7acuxmfpgpptf2sxxj7qc7xm8gzhnklf0sz7u90d6z6x4vj
md1f5n158pxh36ehykyezegzrvq3ltnhcyfpxpcss78u59v8gsu55w3j3crtt4vs33fy3jupswd3pht0yqkvmr
md1f5n158pd6u0qh2z2mwrxfjvd9tg0crh8n4t07049utsul4w5dyeanq0c5j5e0v2yeazrsfwymhg943ez9q
md1f5n158ph3xdp7jdw4x23agzgmyeta0sgkvqq005enx2qvg99s9s4zypw88eyeyvszvszk2mu8es7jcj3
md1f5n158plxfygzdkmhugqfedcfjakelta7k97h5hpszwrfazruxtax3rqls29j4mj2q94ztc62lf0f9p
md1f5n158zpjrhst0rkvn9e775u2nf9sgxcdxq2lqewdp0jr30af03vhes8kuf8wsfhhs7qxv0acr9nj92n
md1f5n158z2d24tpdkh3prfu747g4nt2488aal33v7xgtn7zyxeyt99rm5cmeavkayw4e0s24a9vcq6xq8de
md1f5n158z4nmhnnfk2hd56yhtk4l277vguwcqqaqx5vfn7xcy7t7l04mrquh9k7vj0bywqmpddyhf20urlg
md1f5n158z6t25tltp3ckcyejn3mjr868cj73pd0pw3euqyg9glwyfhcscvvp9u8hdqmfh3syex2yyjd4kn6f
md1f5n158rxpc8xhu3sqxhwsyzw9jr32lcrjputwmdtlut5elkhtynnq7tm8drts47t2qkqmx3ulr0fwgpk
md1f5n158r2vnc02a9rgcupzytm7nek6p8gwfmfg27uwjrp92q0n77qtrsyu48ualq5q0vsgjq0c8zssp3w30
md1f5n158r3ujfumqcaemfyf3cqu633yeh80klhqknfyd4kvsjnnw05ng0cl9h7z2z2ecw
warning: --experimental relaxed the signature rule. This descriptor has at least one spend path that needs NO key, so whoever
r learns its preimage can spend it alone. If that preimage is engraved, THE PLATE IS BEARER ACCESS. Malleability, resource l
imits, repeated keys and timelock mixing were still checked.
note: stdout is watch-only — public keys only, cannot spend
```

```
[exit 0]
```

The keyed card set is 15 md1 chunk(s).

```
wallet-descriptor-template-id (keyless): daee67be4eacf85e8b832ae64fc06566
wallet-descriptor-template-id (keyed)  : daee67be4eacf85e8b832ae64fc06566
wallet-policy-id                   (keyed) : 174ee71427fcb8edcec5d14ae5989449
```

ONE WALLET, TWO IDS. The template id is key-STABLE and identical either way; the policy id depends on the keys. A card binds to whichever of the two matches its own form.

wsh — descriptor and addresses

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md descriptor md1f5n158qpp2040veppfx8qqpx2v6aahst9z5qtlk2pxnhcj
8c9w4uzfy0x4ggr0ff7jgeh2qcmjtmnm68 md1f5n158qdg06453fusycggzn9xdwqqqqpxdweecy63hcar79u3zfygx57kvxs5x9uqf93jqxfl9ppns md
1f5n158qhmwmqmkran74xxmyryu7mk45vzzuv5e4kqq3aaggz4fntvqp28rqw5ws7hc37lyqg66puucsn7wgk md1f5n158qm5578mfalmzaps9gsnxvyuyxlla7a
cuxmfpgptf2sxxjqu7cxmw8gzhnklf0sz7u90d6z6x4vj md1f5n158pxh36ehykyezegzrvq3ltnhcyfxpvcss78u59v8gsu55wj3crtt4vs33fy3jupswd3pht
0yqkvmr md1f5n158pd6uq0h2z2mwrjxjfdv9tg0crh8n4t07049utsul4w5dyeanc0c5j5e0v2yeazrsfwymhg943ez9q md1f5n158ph3xdp7jdw4x23agzgmt
yeta0sgkvqq005enx2qvg99s9s4zypw88eyeyvszvszk2mu8es7jcj3 md1f5n158plxfygzgdkmhugqfedcfjakelta7k97h5hpszwrfazruxtax3rqls29j4
mj2q94ztc62l1f0f9p md1f5n158zpjrhst0rkv9m9e77su2nf9sgxcdxq2lqewdp0j30af03vhes8kuf8wsfhhs7qxv0acr9njq92n md1f5n158z24tpdkh3p
rfu7f47g4ntt488aal33v7xgtm7nzyxeyt99rm5cmeavkayw4e0s24a9vcq6xq8de md1f5n158z4nmhnfk2h56yhptk4l1f27gvugwccqqaqx5vfn7xycy7t7l04m
rquhq5k7vj0yqwmrddyhf20urlg md1f5n158z6t25ltlp3ckcyejn3mj9r868cj73pd0pw3euqyg9glwyfhcsvgsp9u8hdqmfh3syex2yyjd4kn6f md1f5n158r
xpc8xhhu3sqxhwsyzw9jr32lcrjputwmdtlut5elkhtynnnq7tm8drts47t2qkqmx3ulr0fwgpkmd1f5n158r2vnzc02aqrqcupzytm7nek6p8gwvmfmg27uwjrp
92q0n77qtrsyu48ualq5q0vsjq0c8zssp3w30 md1f5n158r3ujfumqccaemfyf3cqu633yeh80klhqknfyd4kvjsnwn05ng0clf9h7z2z4ecw
wsh(or_i(and_v(v:sha256(ede0b28a805feca09a77c48f82babcc1249c79aa840de94fa4a85b5f55a453c813),multi(3,[39ec1b6e/270028'/0'/9'/
1']xpub661MyMwAQrBcGxALkZqYHiTZg3UgzcdVxNzbCBdBjT2ViaCZYexPy2fzQnVfu7Lw5DFYpmrTsp9nXydScuJhD3VEEJhhdoEDw2rEmq3qQ8s/<0';1>/*,
[d02bcdedf/270028'/0'/9'/1']xpub661MyMwAQrBcGsZt7ZRaPrRbPnVojL35MWzsbBrXyMwtbrBLbr3yE2qeffvCXq9tkyqXAidxdgtYYkiVVDHtyB9tdJMJ5
MFMs8'TfnP92Jc/<0';1>/*,[5fd78eb3/270028'/0'/9'/1']xpub661MyMwAQrBcGv2wz7ULSPNpQJbCYTLzd5mZPVZPwphrYvTcQK6tgP5LLPtKdCfYYNdF0
1f1fc61HeTZ6Zv9dw9KMuq8WYZ7GVCEzBoX9f/<0';1>/*),or_i(and_v(v:older(32768),and_v(v:sha256(9ce09a8df1d1f8bc8892441a9eb30d0a18b
c7db81bb0fb3f54c6d9064e7b76ad),multi(2,[92c4c8b2/270028'/0'/9'/1']xpub661MyMwAQrBcFYVZlCYQR8QLuWzVXyGcgJw6VRM8oaauwKFP94sa3h
ftUjQjjZtg7gqlcYRizTdk1mYtb7A2P5gUYyA9CBX8pg1pb8x5amq/<0';1>/*,[086c047e/270028'/0'/9'/1']xpub661MyMwAQrBcFJDpm1NKXvsQyiXHWHy
JXsx6rXViuwj7r6F55PqpCHdWnst2YGBPrwYCapMzqA82mCBdw8BWyTbUfdaw1JawqTrUnNtt1Mq/<0';1>/*)) ,or_i(and_v(v:after(1173520),pk([cfec8
224c/270028'/0'/9'/1']xpub661MyMwAQrBcFhReYzdHhyfKEAgnn7MxQptt1gEAgQz5gQZXwVt7hTA9nSicshD5hCGc5b4wj38gSuR7jAt4gEgUwJZnc7ji
cbbwzMd5a/<0';1>/*)) ,and_v(v:after(1383520),sha256(4743d7c47df21d29e3ed3dfec5d0c0a884ccc2708637ddd771c36d214056954))))#aqe
ftg67
note: stdout is watch-only – public keys only, cannot spend
[exit 0]
```

1243 bytes

Five receive and five change addresses, on the host (5 and 5 derived):

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md address md1f5n158qpp2040veppfx8qqpx2v6aahst9z5qtlk2pxnhcj8c9
w4uzfy0x4ggr0ff7jgeh2qcmjtmnm68 md1f5n158qdg06453fusycggzn9xdwqqqqpxdweecy63hcar79u3zfygx57kvxs5x9uqf93jqxfl9ppns md1f5
n158qhmwmqmkran74xxmyryu7mk45vzzuv5e4kqq3aaggz4fntvqp28rqw5ws7hc37lyqg66puucsn7wgk md1f5n158qm5578mfalmzaps9gsnxvyuyxlla7acux
mfpgptf2sxxjqu7cxmw8gzhnklf0sz7u90d6z6x4vj md1f5n158pxh36ehykyezegzrvq3ltnhcyfxpvcss78u59v8gsu55wj3crtt4vs33fy3jupswd3pht0yq
kvmr md1f5n158pd6uq0h2z2mwrjxjfdv9tg0crh8n4t07049utsul4w5dyeanc0c5j5e0v2yeazrsfwymhg943ez9q md1f5n158ph3xdp7jdw4x23agzgmyet
a0sgkvqq005enx2qvg99s9s4zypw88eyeyvszvszk2mu8es7jcj3 md1f5n158plxfygzgdkmhugqfedcfjakelta7k97h5hpszwrfazruxtax3rqls29j4mj2
q94ztc62l1f0f9p md1f5n158zpjrhst0rkv9m9e77su2nf9sgxcdxq2lqewdp0j30af03vhes8kuf8wsfhhs7qxv0acr9njq92n md1f5n158z24tpdkh3prfu
7f47g4ntt488aal33v7xgtm7nzyxeyt99rm5cmeavkayw4e0s24a9vcq6xq8de md1f5n158z4nmhnfk2h56yhptk4l1f27gvugwccqqaqx5vfn7xycy7t7l04mrqu
hq5k7vj0yqwmrddyhf20urlg md1f5n158z6t25ltlp3ckcyejn3mj9r868cj73pd0pw3euqyg9glwyfhcsvgsp9u8hdqmfh3syex2yyjd4kn6f md1f5n158rxc
8xhhu3sqxhwsyzw9jr32lcrjputwmdtlut5elkhtynnnq7tm8drts47t2qkqmx3ulr0fwgpkmd1f5n158r2vnzc02aqrqcupzytm7nek6p8gwvmfmg27uwjrp92q
0n77qtrsyu48ualq5q0vsjq0c8zssp3w30 md1f5n158r3ujfumqccaemfyf3cqu633yeh80klhqknfyd4kvjsnwn05ng0clf9h7z2z4ecw --chain 0 --cou
nt 5
bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kyculcgz3dwqf266zj
bc1qr6357aaaru5m82sde0xta75q6nj29gsd3m5krvxl1lekq89fs53lqwxkwz4
bc1q3774tl4ntpw0t842cdxtw2ljy82s0c47see7lujg5hs0zccqkjs6tj1w4
bc1qshj4vra0cdcautnksujvh327dtxn394e2npvx6lcrdc4vrrhcv5qf6ga0u
bc1qyld70fxhz9x3mn2h5d58yal2hcgpn2na0a8d8pzsh0qd6lucnw2qs3serj
note: stdout is watch-only – public keys only, cannot spend
[exit 0]
```

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/debug/md address md1f5n158qpp2040veppfx8qqpx2v6aahst9z5qtlk2pxnhcj8c9
w4uzfy0x4ggr0ff7jgeh2qcmjtmnm68 md1f5n158qdg06453fusycggzn9xdwqqqqpxdweecy63hcar79u3zfygx57kvxs5x9uqf93jqxfl9ppns md1f5
n158qhmwmqmkran74xxmyryu7mk45vzzuv5e4kqq3aaggz4fntvqp28rqw5ws7hc37lyqg66puucsn7wgk md1f5n158qm5578mfalmzaps9gsnxvyuyxlla7acux
mfpgptf2sxxjqu7cxmw8gzhnklf0sz7u90d6z6x4vj md1f5n158pxh36ehykyezegzrvq3ltnhcyfxpvcss78u59v8gsu55wj3crtt4vs33fy3jupswd3pht0yq
kvmr md1f5n158pd6uq0h2z2mwrjxjfdv9tg0crh8n4t07049utsul4w5dyeanc0c5j5e0v2yeazrsfwymhg943ez9q md1f5n158ph3xdp7jdw4x23agzgmyet
a0sgkvqq005enx2qvg99s9s4zypw88eyeyvszvszk2mu8es7jcj3 md1f5n158plxfygzgdkmhugqfedcfjakelta7k97h5hpszwrfazruxtax3rqls29j4mj2
q94ztc62l1f0f9p md1f5n158zpjrhst0rkv9m9e77su2nf9sgxcdxq2lqewdp0j30af03vhes8kuf8wsfhhs7qxv0acr9njq92n md1f5n158z24tpdkh3prfu
7f47g4ntt488aal33v7xgtm7nzyxeyt99rm5cmeavkayw4e0s24a9vcq6xq8de md1f5n158z4nmhnfk2h56yhptk4l1f27gvugwccqqaqx5vfn7xycy7t7l04mrqu
hq5k7vj0yqwmrddyhf20urlg md1f5n158z6t25ltlp3ckcyejn3mj9r868cj73pd0pw3euqyg9glwyfhcsvgsp9u8hdqmfh3syex2yyjd4kn6f md1f5n158rxc
8xhhu3sqxhwsyzw9jr32lcrjputwmdtlut5elkhtynnnq7tm8drts47t2qkqmx3ulr0fwgpkmd1f5n158r2vnzc02aqrqcupzytm7nek6p8gwvmfmg27uwjrp92q
0n77qtrsyu48ualq5q0vsjq0c8zssp3w30 md1f5n158r3ujfumqccaemfyf3cqu633yeh80klhqknfyd4kvjsnwn05ng0clf9h7z2z4ecw --chain 1 --cou
nt 5
bc1q70d7dmaz2s4ur98vewnzyuuct6wzu3jpwihhn8dz0g5agkevzjaqnn6ecs
bc1qgw504j2sphtt3rpy4hthwgnlgm3fyv8h7pc8nzeq5k68z6h75kzs35gn79
bc1qm7dxt9ghlf3nh5ump80vhyL9123txpp5qqjcun2s29xcgqfjwndqgcwr42
bc1qsgqp0dwmatl72kdkj3gnhkwqrq9seu3pfpshlwtmnam82ktdp37qflqt8p
bc1qdmvtv44gaj5e77apzkz9gfrhnrtrtn5nmn86qts8uw7mnmkwxkusvve7kd
note: stdout is watch-only – public keys only, cannot spend
[exit 0]
```

The key cards bind to the TEMPLATE id

An mk1 card carries a `policy_id_stub`, and the stub is **form-aware**: a card minted against the keyed policy roots on the wallet-policy id, while one minted against the keyless template roots on the template id. Same wallet, two ids — so a card made for the wrong form is refused at membership before seating is even attempted.

```
$ cat /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/keys.txt
[39ec1b6e/270028'/0'/8'/0']xpub6Dn2FdZCkR6b6nCFWKMZraCwEcC7aedkRrVHf8DK5gSBry6YJkiUZp9bL8j2a4osHSuhDsk1FBnSS81C82qvZ4uiDofJ
GQKmNmmxqfUxCr
[d02bcedf/270028'/0'/8'/0']xpub6DdqV8W2dzVkyfWmNUJHkWoDh6JctbyGvKrJVUDUL1hSoUBGtfB3JtXoyYtvkVR3S6jyzagALehL56vqwcjXAc5hoaCoL
6yyrtCLQf1dg2f
[5fd78eb3/270028'/0'/8'/0']xpub6FMyX6RDPGTJgt3hYThGymmhRUhntuA9HagmHmDPum11hDKk6HLNFspSyRWqqdiSQYMQujjsL2Xn9ERJE9YDZR9jG2X4U
Jbx53XE3b8Z5Bw
[92c4c8b2/270028'/0'/8'/0']xpub6E9jU3J1VtuNb5GVBatKMHYkHiMM7dD6yK2x4fhDDSGhXHLxJQgFZnwLEXwc1jAb5ZCsNUhQGDqUdy3Jt3kRmSMG6Fny
BnM1PRXmD6bXb4
[086c047e/270028'/0'/8'/0']xpub6Ep7XhwjqPzjf5HTCQTf8M8QRYUqGHJLPxQ9qbU7xJrGGCsG9Fba4QUtughDTBLJgf5praqaxKYUhmTvKMsWiG8Gs24jJ
gLzspG2tstkBdz
[cef8224c/270028'/0'/8'/0']xpub6EvqkLWC3QnMavFkcHB5Kzfnv8FeikdNwurio9zyTWgNnwsZXf7KdcGAh9to36zwLckiVklpyjBUxD9f9xwSuT9KUmw1
v1QzqDuYuBw7q
[exit 0]

note: stdout is watch-only – public keys only, cannot spend
$ /scratch/code/shibboleth/mnemonic-engrave/design/journeys/mk_cards_from_json.py /scratch/code/shibboleth/mnemonic-engrave/
design/journeys/out/rcw/tr/mk-encode.json /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/mk-encode-ra
w.txt
6 key cards -> 18 mk1 chunks
[exit 0]

template-id prefix: 68a1a888   stub the cards embed: 68a1a888
```

The transcript gates on that equality rather than printing it: if the cards ever bound to the policy id instead, the run goes red.

Two wallets, and the proof they are two

wrapper	template-id	first receive address
tr	68a1a888385797337ce5debc90fcfb1e	bc1p8h77a4mgl49mld6crpdpvx4js6gclkv9vr93enl54cdz0gp6v8smuqr2n
wsh	daee67be4eacf85e8b832ae64fc06566	bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kycu1cgz3dwqf266zj

Different ids, different addresses, as they must be.

This is gated, not observed. If a future edit ever made the two wrappings produce the same template id or the same addresses, the transcript fails — because at that point one of them is not the wallet the document says it is.

The device half — four arms

The same cards carried into the SeedHammer II emulator over NFC, by `capture_rcw.py`. Two routes per wrapper, and the second is the one that matters:

- **keyed** — the 15-chunk card alone. It carries its own keys, so nothing is seated. Proves the device can derive this wallet at all.
- **seating** — the 5-chunk keyless template plus its six mk1 cards, which is *what actually gets engraved*. Proves the device can join them.



tr / seating: 5 template chunks assembled, 6 key cards gathered.

tr / seating: the addresses, seated from the key cards.

tr — both routes agree

on the device AND on the host
bc1p8h77a4mgl49mld6crpdpvx4jsg6gc1kv9vr93en154cdz0gp6v8smuqr2n
bc1panm4lv4h18n96xccufjc7rkd4hhuv1ed2hp5eqdhncaqvpuvvfdq5z9t8t
bc1pt7g96plv4mw73vis64wvq2eswq9zrh4r0ztnx7huuma94xzskkhqm8vkn4
bc1pp3fpcxm9zpk75hguhj5qh804wxzrff5tp6wcunr0up7j3xp46v3su0mntq

Keyed and seating produced **the same** addresses.

wsh — both routes agree

on the device AND on the host
bc1qr6h5gahcaqa8a35p3ts0d2w6qvhmsn7dhunu5xd9kyculcgz3dwqf266zj
bc1qr6357waaru5m82sde0xta75q6nj29gsd3m5krvx1lekq89fs53lqwxkwa
bc1q70d7dmaz2s4ur98vewnzyuuct6wzu3jpwhhhn8dz0g5agkevzjaqnn6ecs
bc1qqw504j2sphtt3rpy4hthwgnlgm3fyv8h7pc8nzeq5k68z6h75kzs35gn79

Keyed and seating produced **the same** addresses.

That equality is the whole point.
One wallet reached two ways — from a card that carries its keys, and from a template plus six separate key cards — across the air gap, through two implementations in two languages. Either route alone would only prove the device is self-consistent.

Five on the host, two on the device

The host derives five addresses per chain. The device shows **two**, and that is deliberate: `addrProofPerChain = 2` (`gui/wallet_policy.go:184`), chosen by plan D6 as *"receive AND change, FEWER of each — change is where a policy mismatch silently loses funds, so proving both chains derive beats proving one chain five times."*

Asked to reconsider for this journey, the ruling was to leave it (`design/agent-reports/RULING_sh2_address_count.md`). The argument: **every realistic mismatch diverges at index 0**, and an index-conditioned backdoor defeats a five-address screen exactly as easily as a two-address one. The extra three addresses buy reading fatigue on a *consent* surface and no additional evidence.

So the device proves a

sample

and the host proves the **range**, and the walk asserts the device's four are byte-equal to the host's indices 0–1 per chain — rather than quietly comparing fewer and calling it agreement.

The comparison can fail

An assertion that addresses *appear* would pass for a walk that silently stopped driving the device. `--prove-it-can-fail` corrupts one character of one expected address and requires the walk to refuse; it exits 0 only when the walk fails.

Getting it to the device: the payload route

`me sysw pack` builds the SYSTEMWIDE container the device reads. This journey's wallet carries no ms1 secret share -- the three preimages are engraved as text, not as ms1 -- so the payload holds only public strings.

Strings for the payload: 46 total, ms1 count = 0

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me sysw pack --in /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/payload-strings.txt --out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/payload.bin --no-passphrase
strength: no passphrase — BELOW the threshold
digest:   be81 c617 9403 3c05 0287 480a af95 5d44
runcap: CAPTURE FAILED -- no stdout line matched ./
[exit 1]
```

Payload: 3887 bytes.

```
$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me sysw show /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/payload.bin
sealed:   false
pub_len: 3835
ct_len:   0
identity: 04ca5f8938a7df7897af8a169ad9bc0595dfeaa93815b9a5b9cce402eff7f06f
digest:   be81 c617 9403 3c05 0287 480a af95 5d44
public record 0: mdl/mk1 — confirmed
public record 1: mdl/mk1 — confirmed
... 41 lines elided ...
public record 43: mdl/mk1 — confirmed
public record 44: mdl/mk1 — confirmed
public record 45: mdl/mk1 — confirmed
[exit 0]
```

public records in the container: 46 strings fed in: 46
... clipped; full text in transcript_rcw.txt

Gated both ways: the container's record count must equal the number of strings fed in, and the ms1 count must be **zero**

— this wallet's three passphrases are engraved as text, not as ms1 shares.

The sealed route, and the limit it has

```
`me seal` encrypts the records and emits a UF2 for `picotool load`. The
passphrase is GENERATED, never supplied -- a memorable one does not survive an
offline attack on a stolen machine.

IT IS SEALED PER WRAPPER, AND NOT BY CHOICE. `me seal` accepts 1..=24 records;
the two bundles together are 46, so the combined set is REFUSED. Each wrapper's
23 strings fit with one to spare. That is a real limit of this route and the
unsealed `sysw pack` above does not share it -- it took all 46.

$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me seal --in /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/payload-strings.txt --out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/sealed-both.uf2 --iterations 100000
me: 46 records; must be 1..=24 (2-of-3 multisig is 15)
[exit 4]

Refused, as documented. Now per wrapper:

-- tr: 23 records
$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me seal --in /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/backup-strings.txt --out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/sealed.uf2 --iterations 100000
me: wrote 4096 bytes to /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/sealed.uf2

passphrase – write this down and store it APART from the machine:

    exhibit gallery crisp illegal inspire jewel winner ethics pluck blast mask vivid

load: picotool load --verify /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/tr/sealed.uf2 (machine in BOOTSEL)
wipe: picotool erase -r 0x10E00000 0x10E10000
runcap: CAPTURE FAILED -- no stdout line matched ./
[exit 1]

    sealed UF2: 4096 bytes

-- wsh: 23 records
$ /scratch/code/shibboleth/mnemonic-engrave/target/debug/me seal --in /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/wsh/backup-strings.txt --out /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/wsh/sealed.uf2 --iterations 100000
me: wrote 4096 bytes to /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/wsh/sealed.uf2

passphrase – write this down and store it APART from the machine:

    hour speak abandon curtain round grass achieve latin result urban quick half

load: picotool load --verify /scratch/code/shibboleth/mnemonic-engrave/design/journeys/out/rcw/wsh/sealed.uf2 (machine in BOOTSEL)
wipe: picotool erase -r 0x10E00000 0x10E10000
runcap: CAPTURE FAILED -- no stdout line matched ./
[exit 1]

    sealed UF2: 4096 bytes
```

The two routes do not have the same capacity.

`me seal` accepts 1..=24 records; both bundles together are 46, so the combined set is **refused**. Each wrapper's 23 fit, with one to spare. The unsealed `sysw pack` took all 46 without complaint.

The transcript demonstrates that refusal and then gates on it still happening. If the cap is ever raised, that gate goes red and this page gets rewritten — rather than quietly becoming false.

What this journey does NOT show

- **No real hardware.** The device half is the emulator — the same Go firmware compiled to wasm, driven over the same NFC entry point, but not a machine with a screen and a hammer.
- **Nothing was engraved.** The plates are rendered previews. No steel was cut, and the payloads were built but not flashed to a device.
- **No wallet-file export.** Nunchuk, Sparrow and Bitcoin Core outputs are a separate piece of work; at the time of writing no export surface exists anywhere in the constellation. The Sparrow answer is already known to be **no**, and worse than no — see [design/agent-reports/PLAN_export_sparrow.md](#).
- **No spend.** Every address here is derived, none is funded, and no tier has been exercised against a real chain. The timelocks in particular are arithmetic in this document and consensus rules everywhere else.